Suman Maiti

PhD Student, Department of Computer Science and Engineering, IIT Kharagpur

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Education

Indian Institute of Technology Kharagpur Kharagpur, India
Doctor of Philosophy (PhD) in Computer Science and Engineering *Jan 2022 – Present*

- **Research Focus:** Cyber-Physical Systems Security, Smart Grid Security, Formal Methods, Safe RL.
- **Coursework:** High Performance Computer Architecture, Foundations of Algorithm Design and Machine Learning, AI for Cyber-Physical Systems, Computational Foundations of CPS, Statistical Methods.

Kalyani Government Engineering College Nadia, India
Bachelor of Technology (B.Tech) in Electrical Engineering | ***DGPA: 9.03/10*** *Sep 2017 – Aug 2021*

D.A.V. Model School, IIT Campus Kharagpur, India
All India Senior School Certificate Examination (Class XII) | ***93.8%*** *Mar 2017*

Research Experience

Junior Research Fellow (JRF) Oct 2021 – Dec 2023
Indian Institute of Technology Kharagpur *Kharagpur, India*

- **Smart Grid Privacy and Security:** Conducted research into vulnerabilities in smart grid environments and their countermeasures.
- Applied machine learning techniques, particularly reinforcement learning, to identify vulnerabilities in IEEE 9, 14, 37, and 39 bus power grid models.
- Investigated industrial SCADA communication protocols including DNP3, DNP3-SA, and Modbus for threat modeling.

Summer Research Intern Jun 2019 – Jul 2019
Indian Institute of Technology Kharagpur *Kharagpur, India*

- **Design of a Digital Clock using 8085 Microprocessor:** Implemented microprocessor programming to display time using a seven-segment display, gaining foundational experience in low-level hardware-software interaction.

Selected Projects

Formal Methods for Smart Grid Privacy and Security | *Staliro, RL, IEEE Bus Models*

- Investigated vulnerabilities of IEEE 9, 14, and 39 bus power grid generation and distribution units using a reinforcement learning agent combined with the formal falsification tool Staliro.
- Produced targeted attack sequences and formal safety proofs under Metric Temporal Logic (MTL) specifications.

Fault Studies of Wind-Farm Embedded Power Network | *MATLAB/Simulink*

- Simulated symmetrical and asymmetrical fault conditions in a 225/90 kV power transmission network.
- Analyzed voltage and power waveforms of transmission lines during fault conditions (Published at IEEE DevIC 2021).

CPU Architecture Performance Estimation | *GEM5 Simulator, C++*

- Developed a framework for generating different CPU configuration combinations from user-specified parameters.
- Evaluated performance by running benchmark programs on the GEM5 system simulator, enabling systematic architectural trade-off analysis.

Autonomous Driving in F-1/10 Vehicle | *Control Theory* | *In Progress*

- Designing a controller for an F-1/10 autonomous vehicle that enables it to follow a given track while avoiding obstacles with formal safety guarantees.

Publications

Journal Articles

- S. Ghosh, **S. Maiti**, D. Mukhopadhyay, S. Dey. *"Differentially Private Real Time Pricing Control for Smart Grids"*, ACM Transactions on Cyber-Physical Systems, 2025.

Conference Proceedings

- **S. Maiti**, A. Balabhaskara, S. Adhikary, I. Koley, S. Dey. *"Targeted Attack Synthesis for Smart Grid Vulnerability Analysis"*, Proceedings of the ACM SIGSAC Conference on Computer and Communications Security (CCS '23), 2023. **[Cited by 18]**
- **S. Maiti**, S. Som, S. Mondal, P. K. Gayen. *"Symmetrical and Asymmetrical Fault Studies of Wind-Farm Embedded Power Network"*, Devices for Integrated Circuit (DevIC), IEEE, 2021.

Preprints

- **S. Maiti**, S. Adhikary, S. Dey, A. R. Hota. *"Learning-Enabled Adaptive Voltage Protection Against Load Alteration Attacks on Smart Grids"*, arXiv:2411.15229, 2024.
- **S. Maiti**, S. Dey. *"Smart Grid Security: A Verified Deep Reinforcement Learning Framework to Counter Cyber-Physical Attacks"*, arXiv:2409.15757, 2024. **[Cited by 11]**

Teaching Assistant Experience (IIT Kharagpur)

Programming and Data Structures Laboratory — Jan 2026 – May 2026
Spring Semester 2026 | Instructor: Prof. Dipanwita Roy Chowdhury

Reinforcement Learning — Jul 2025 – Jan 2026
Autumn Semester 2025 | Instructor: Prof. Soumyajit Dey

Formal Language and Automata Theory — Feb 2025 – Jul 2025
Spring Semester 2025 | Instructors: Prof. Soumyajit Dey & Prof. Animesh Mukherjee

Programming and Data Structures Theory — Jul 2024 – Jan 2025
Autumn Semester 2024 | Instructor: Prof. Soumyajit Dey

Formal Language and Automata Theory — Aug 2023 – Mar 2024
Autumn 2023 – Spring 2024 | Instructors: Prof. Soumyajit Dey & Prof. Animesh Mukherjee

Programming and Data Structures Laboratory (3 Semesters) — Oct 2022 – Jul 2023
Instructors: Prof. Pallab Dasgupta, Prof. Soumyakanti Ghosh, Prof. K. S. Rao, Prof. Debasis Samanta

Professional Service & Mentorship

Professional Service & Peer Review

- **External Reviewer (Journals):** IEEE Transactions on Power Systems, IEEE Transactions on Smart Grid, IEEE Transactions on Industrial Cyber-Physical Systems, IEEE Transactions on Computer-Aided Design of Integrated Circuits & Systems (TCAD).
- **External Reviewer (Conferences):** ICCPS (2024, 2025, 2026), ICCPS Artifact Evaluation Committee (2026), AAAI (2023), Euromicro Conference on Digital System Design (DSD) (2023, 2024), IEEE Real-Time Systems Symposium (RTSS) (2023).

Mentorship & Academic Services

- **AICTE Faculty Quality Improvement Program:** Guided 5 faculty members on the application of Reinforcement Learning in Cyber-Physical Systems Operation.
- **Student Mentorship:** Mentoring B.Tech and Dual Degree thesis students (CSE, IIT Kharagpur) on learning-assisted methods for smart grid vulnerability analysis.

Awards, Grants & Talks

ACM IARCS Travel Grant: Awarded INR 1,00,000 in recognition of research contribution.

ACM CCS 2023 Student Travel Grant: Recipient of travel grant for the ACM Conference on Computer and Communications Security.

CHANAKYA PhD Fellowship (2024-2025): Awarded the prestigious CHANAKYA PhD fellowship by the AI4ICPS Hub, IIT Kharagpur.

Invited Talk: Delivered talk at ACM ARCS Event 2023 at NISER Bhubaneswar on smart grid security and attack synthesis.

GATE 2021 (Electrical Engineering): Secured **94.4 Percentile** in the Graduate Aptitude Test in Engineering, qualifying for doctoral admissions.

Technical Skills

Programming Languages: Python, C++, C, PHP, LaTeX
ML / DL Frameworks: TensorFlow, Keras, Scikit-learn, Stable Baselines3, Gymnasium, PyTorch
Formal Verification: Z3 SMT Solver, Staliro (Falsification), UPPAAL, TuLiP, Metric Temporal Logic, Hybrid Automata
Engineering Tools: MATLAB, Simulink, RT-Lab, LabVIEW, Artemis, HYPERSIM, EMTP-RV / Ephasorsim
Hardware & Platforms: OP4510 (Opal-RT), SEL-421 (Schweitzer), Raspberry Pi, Linux, Windows
SCADA & Protocols: DNP3, DNP3-SA (Secure Authentication), Modbus RTU/TCP
Development: Git, Docker, Jupyter Notebook, pandapower, GEM5 Simulator